

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - II

Roll No. : S033	Name: Rudra Rawal
Class : SYIT	Batch : 1
Date of Assignment : 2026-07-22	Date/Time of Submission : 2026-07-22

Questions:

2a. Write a python code to create nested tuples.

CODE:

```
Practical_2A.py - C:/Users/mvlui/Desktop/SYIT_033/PDS/Practical_2A.py (3.14.6)
File Edit Format Run Options Window Help
students = ("Rudra", 33, ("BscIT", "Semester III"), 90)

print("Name: ", students[0])
print("RollNo.: ", students[1])
print("Department: ", students[2][0])
print("Sem: ", students[2][1])
print("Percentage", students[-1])

print("\nRudra S033")
```

OUTPUT:

```
Name: Rudra
RollNo.: 33
Department: BscIT
Sem: Semester III
Percentage 90

Rudra S033
```

2b. Write a python code to sort the nested tuple using sorted () function.

CODE:

```
tup = ((1,3), (2,5), (4,2), (5,3))

Sorted_data = sorted(tup, key = lambda x: x[1])
print(Sorted_data)
```

OUTPUT:

```
[(4, 2), (1, 3), (5, 3), (2, 5)]

Rudra S033
```

```
Traceback (most recent call last):
  File "C:/Users/mvlui/Desktop/SYIT_033/PDS/Practical_2A.py", line 2, in <module>
    tupl[1] = (4,3)
TypeError: 'tuple' object does not support item assignment
```

Curiosity Questions:

Q1. Define a tuple. How is it different from a list? Give one example.

Q2. Write a Python program to create the following tuples:

- a) An empty tuple
- b) A tuple with integers only
- c) A tuple with mixed data types
- d) A nested tuple (tuple inside another tuple)

Q3. Write a program to access elements of a tuple using indexing and slicing.

Q4. Create a tuple and find the following:

- Length of the tuple
- Maximum and minimum value (if all are numbers)
- Count of a specific item

Q5. Write a program to convert a list into a tuple.

Q6. Write a program to sort a tuple of numbers in ascending order.

Q7. Write a program to show that tuples are immutable (cannot be changed).

Q8. Write a program to clone (copy) a list and print both original and copied lists.

Q9. Write the output of the following code:

```
t = (1, 2, 3, 4, 5)
print(t[1:4])
```

Q10. Create a tuple of student marks: (78, 85, 92, 88, 76). Write a program to:

- Find average marks
- Display marks above 80